Jointly Optimize Throughput and Localization Accuracy: UAV Trajectory Design for Multiuser Integrated Communication and Sensing

Siyan Gu, Chunbo Luo, *Senior Member, IEEE*, Yang Luo, and Xiaoguang Ma

***Abstract*—Unmanned aerial vehicle (UAV) is becoming a crucial aerial platform to provide emergency or enhanced communication and sensing services benefited from its unique features, including agile mobility and high probability of Line-of-Sight coverage. In this article, we investigate the novel UAV trajectory design problem where the UAV communicates with multiple users and simultaneously senses the positions of multiple targets, via the integrated communication and sensing design. To evaluate the overall system utility, we first derive the communication throughput and localization error Cramér–Rao bound (CRB) model and highlight the coupled challenge and tradeoffs on UAV's trajectory. We thus model the three typical integrated sensing and communication scenarios: 1) communication centric; 2) sensing centric; and 3) tradeoff scenarios. We further introduce path discretization to support in-flight communication and hovering sensing, which make it possible to jointly optimize the UAV trajectory, communication throughput and target localization estimation error CRB with limited complexity. Because this optimization problem involves integer programming caused by multiuser association, we propose a novel trajectory initialization framework based on traveling salesman problem to determine the service order for sensing targets and the initial UAV trajectory. To address the nonconvexity of this optimization problem, we propose an efficient iterative algorithm using the successive convexity approximation technique to obtain the approximate optimal solution, which is dynamically reconfigured and optimized with updated information in flight. Extensive numerical results demonstrate that the proposed algorithm achieves superior performance in the tradeoff between average achievable rate and CRB in all three scenarios.**

***Index Terms*—Cramér–Rao bound (CRB), integrated sensing and communications (ISAC), unmanned aerial vehicle (UAV).**

I. Introduction

THE UTILIZATION of unmanned aerial vehicles (UAVs) for wireless communication and sensing can facilitate wireless coverage and sensing data acquisition in areas lacking adequate or ground-based infrastructure, particularly for post-disaster scenarios or temporary high-demand outdoor areas [1]. Due to the characteristics of on-demand deployment, high mobility, and robust Line of Sight (LoS) links connectivity inherent to UAVs, they are flexibly employed across numerous application domains, such as intelligent transportation and emergency communications, holding high potentials in the future development of the Internet of Things (IoT) [2]. First, due to the deployment of UAVs at higher altitudes, in comparison to traditional ground-based communications, UAV networks can offer highly probable LoS air-to-ground channels to enhance communication quality. This is equally advantageous for sensing applications, as sensing performance is severely compromised under Non Line-of-Sight (NLoS) links [3]. On the other hand, due to the fully controllable mobility, UAVs can be deployed in optimal locations, and also dynamically scheduled as mobile access points (APs) around communication users (CUs) or sensing targets to reduce signal transmission power and enhance network performance. In recent years, extensive researches have been conducted in the field of UAV-enabled wireless communication networks. In [4], the UAV sequentially hovers over user clusters to provide services. Zeng et al. [5] has considered the energy minimization network involving communication between the rotary-wing UAV and multiple users. Furthermore, the uniqueness of UAVs has opened new perspectives for sensing systems in detection, localization, and other aspects. Van Nguyen et al. [6] proposed a UAV online path planning algorithm with joint detection and tracking capabilities. Traditionally, the research in UAV communication and sensing has often been conducted independently, but emerging applications often require both communication and sensing (C&S), the coexistence of the two systems on the same device will lead to spectrum congestion and overloading of UAVs.

5G-and-beyond networks are expected to drive the development of numerous emerging applications that typically demand low latency, high-communication quality, and highly accurate localization, including vehicular networks, autonomous driving, etc. [7]. Within these applications, communication and sensing are indispensable and intertwined. To address these challenges, integrated sensing and communications (ISACs) has emerged as a key technology in the vision of 5G beyond and 6G development [8]. Early research on ISAC systems

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Siyan Gu, Chunbo Luo, and Yang Luo are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: 202221011626@uestc.edu.cn; c.luo@uestc.edu.cn; luoyang@uestc.edu.cn).

Xiaoguang Ma is with the College of Information Science and Engineering and the Faculty of Robot Science and Engineering, Northeastern University, Shengyang 110167, China (e-mail: maxg@mail.neu.edu.cn).

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focus on ISAC waveform design, aiming to design joint waveforms to achieve both target sensing and user communication dual functionalities. The ISAC waveform design is generally categorized into three frameworks: 1) sensing centric; 2) communication centric; and 3) tradeoff designs. Sensing-centric systems have waveform design based on sensing signals, in which communication data are embedded [9], [10], [11], [12]. In Hassanien's et al. [11] pioneering work, the concept of embedding communication data was introduced by altering the sidelobes of radar's spatial beam patterns. These schemes have a primary drawback in that the communication rate is constrained by the pulse frequency of radar waveforms, leading to reduced communication performance. In communication-centric design schemes, classical communication waveforms are used for data transmission and target detection. As the system has knowledge of the communication waveform, it can be employed for single-static detection [13], [14], [15], [16], [17]. However, signal waveforms modulated by communication data will change randomly according to different information transmitted. This may render the sensing performance inadequate to meet the requirements. In view of aforesaid limitations inherent to both systems, recent research trends lean toward a tradeoff design, where the creation of innovative ISAC waveforms facilitates the weighted optimization of C&S, with due consideration to the interplay of mutual interference inherent to these two functions [18], [19], [20], [21], [22], [23]. Liu's et al. [21] research delved into multiuser interference, introduces a branch-and-bound algorithm that achieves an optimal tradeoff between radar and communication performance. Within [22], a general framework was presented to unveil the fundamental tradeoff between C&S. However, these prior works primarily focus on terrestrial ISAC, which have physical limitations. For example, in ground-based ISAC, base stations are often fixed within limited coverage, and the communication channels encounter complex obstacles, leading to severe signal attenuation. This imposes a significant impact to both communication and sensing performance.

Recently, benefiting from the flying height advantages of UAV platforms, the application prospects of the integration of ISAC with UAVs have shown promising perspectives [24]. Wang et al. [25] investigated the maximization of network utility, which was achieved through the optimization of UAV deployment, user association, and transmission power control. While static UAV deployment can enhance the establishment of high-quality communication links by increasing LoS probability, it does not fully exploit the degrees of freedom offered by the mobility. Moreover, dynamic UAV networks can more effectively enhance performance [26], [27]. Meng et al. [26] introduced a novel integrated periodic C&S mechanism to jointly optimize UAV trajectories and beamforming. While the metrics, such as beamforming gains and received signal-to-noise ratios (SNRs), employed in the aforementioned studies, can to some extent gauge system's sensing performance, it does not directly represent the accuracy of target localization. Therefore, in the ISAC networks with target localization requirements, the metrics, such as mean squared error (MSE) and Cramér–Rao bound (CRB) for target localization accuracy, can better reflect the sensing performance compared with the received SNR. Jiang et al. [28] optimized a UAV's trajectory to minimize the weighted sum of the target's PCRB for position and velocity estimates. In [29], a single UAV was considered within a single-user, single-target ISAC system which considers the tradeoff between C&S while optimizing the UAV trajectory. Wu et al. [30] used Kalman filter to predict user's location, focusing on real-time secrecy rate over localization accuracy. In [31], multiple UAVs were introduced to facilitate communication and tracking of a user.

While the prior work has addressed the issue of target localization accuracy, the UAV primarily serves a single user or target, potentially leading to resource wastage and limited efficiency, especially in the common multiuser environments. Moreover, they only consider specific ISAC scenarios and are less general. According to our investigation, research considering multiuser communication and multitarget localization within a single UAV flight has not been reported in the literature. Aiming to address this key challenge, we propose a novel UAV-assisted ISAC model to enable a UAV to serve multiple users and sensing targets, by formulating a joint optimization problem involving the UAV trajectory, communication throughput, and target localization accuracy. The contributions of this article are summarized as follows.

1) We propose a novel UAV-enabled ISAC network model that contains multiple users and multiple sensing targets. The model adopts realistic settings for both communication and sensing tasks, including probabilistic LoS channel model, sensing model, and UAV energy consumption model. We derive the CRB for the coordinates' estimation of the sensing targets based on the echo time-delay measurement model.
2) We suggest a joint optimization framework with in-flight communication and hovering sensing in the three typical ISAC scenarios, and thus formulate the optimization problems to jointly optimize the UAV trajectory, communication throughput and the target localization estimation error CRB, subjecting to constraints related to UAV onboard energy, maximum flight velocity, and flight duration.
3) We introduce a trajectory initialization strategy based on traveling salesman problem (TSP) to determine the service order for multiple CUs and sensing targets and the average probability of LoS links, thereby reducing the complexities associated with integer programming. Additionally, for the nonconvex challenge of the joint problem, we propose an iterative algorithm based on successive convex approximation (SCA), to obtain an approximate optimal UAV trajectory.
4) We introduce a trajectory replanning algorithm to further enhance target localization accuracy, which dynamically reconfigures subsequent trajectories based on the updated information. Extensive numerical results indicate that the proposed algorithms consistently deliver superior communication performance and localization accuracy.

The remainder of this article is organized as follows. Section II introduces the UAV-enabled ISAC system model.

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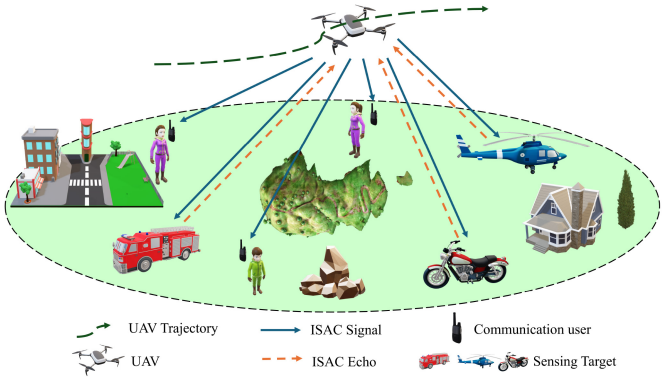


Fig. 1. Considered UAV enabled ISAC system.

Section III formulates the UAV-enabled ISAC problems, while the proposed solutions are derived in Section IV. Our simulation results are provided in Section V, while Section VI concludes this article.

Notations: Unless otherwise specified, matrices and vectors are denoted by bold letters (i.e., $\mathbf{A}$), and scalars are denoted by normal font (i.e., a).We use $\|\cdot\|$ and $[\cdot]^T$ to denote the vector norm and the transposition operation, respectively. We utilize diag$(\cdot)$ to represent a diagonal matrix.

II. System Model

In this section, we consider a UAV-enabled ISAC system shown in Fig. 1, where UAV serves as a base station serving for N CUs, with the set denoted by $\mathcal{N} = \{1, \ldots, N\}$, and K potential sensing targets, with the set denoted by $\mathcal{K} = \{1, \ldots, K\}$. Within the mission time T determined by the onboard energy, the UAV starts from starting point $\mathbf{q}_I$, completes the data transmission and target sensing tasks, and flies to the end point $\mathbf{q}_F$. The UAV flies at a fixed height H, flies or hovers within considered area, transmits signals to N CUs scattered in the area, and performs location estimation on K sensing targets through the echoes reflected by the sensing targets. The CU's location is denoted by $\mathbf{w}_n = [x_n, y_n, z_n]^T, n \in \mathcal{N}$, which are assumed to be known using positioning tools, e.g., GPS. However, K sensing targets are passive targets, which are unknown to the UAV, so their position needs to be estimated through the proposed ISAC system. The target's location is denoted by $\mathbf{p}_k = [x_k, y_k, 0]^T, k \in \mathcal{K}$. Denote by $\mathbf{q}(t) = [x_t, y_t, H]^T$ with $t \in [0, T]$ is the UAV trajectory.

A. Communication Model

For the communication model, let $d_n(t)$ denote the distance from the UAV to the nth CU, $t \in [0, T]$, which is

$$d_n(t) = \sqrt{\|\mathbf{q}(t) - \mathbf{w}_n\|^2}. \tag{1}$$

Therefore, for the link between UAV and nth CU, the channel loss can be expressed as [32]

$$PL_\xi = 20\log(d_n(t)) + 20\log(f) + 20\log\left(\frac{4\pi}{c}\right) + \eta_\xi \tag{2}$$

where ξ refers to the propagation group and $\xi \in \{\text{LoS}, \text{NLoS}\}$. f is the system frequency. c is the speed of light, while η_ξ is the excessive pathloss that has no obvious correlation with the elevation angle between the CU and UAV. The communication links from the UAV to the CUs are dominated by the LoS links. We assume that the Doppler frequency shift caused by UAV mobility is well compensated at the CUs [33] and the sensing receivers [34]. However, complex environments can make communication links obscured, so NLoS links will also be considered. Furthermore, considering the channel combined with LoS and NLoS links, the total average pathloss is

$$\Lambda = \sum_{\xi} PL_\xi P_{n,\xi}(\theta_n(t)) \tag{3}$$

where $P_{n,\xi}(\theta_n(t))$ represents the probability of occurrence of a certain propagation group, which is dependent on the elevation angle $\theta_n(t)$ between the CU and the UAV calculated by $\theta_n(t) = \frac{180}{\pi}\arcsin\frac{H-z_n}{d_n(t)}$. The relation of the occurrence probability of the propagation group as the following:

$$P_{n,\text{NLoS}}(\theta_n(t)) = 1 - P_{n,\text{LoS}}(\theta_n(t)). \tag{4}$$

In general, the LoS probability is contingent upon the density and height of the building within the signal transmission environment. The $P_{n,\text{LoS}}(\theta_n(t))$ can be represented approximately in the form of a simple modified sigmoid (S-curve) function as [32]

$$P_{n,\text{LoS}}(\theta_n(t)) = \frac{1}{1 + C\exp(-D[\theta_n(t) - C])} \tag{5}$$

where C and D are parameters related to the propagation environment. It is clear that the UAV trajectory $\mathbf{q}(t)$ determines the LoS probability $P_{n,\text{LoS}}(\theta_n(t))$. Accordingly, the received power of the nth CU denoted by $P_{R,n}(t)$ can be expressed as

$$\begin{aligned} P_{R,n}(t) = P_T d_n(t)^{-2}\beta_0^{-2}\big(&k_{\text{LoS}}P_{n,\text{LoS}}(\theta_n(t)) \\ &+ k_{\text{NLoS}}P_{n,\text{NLoS}}(\theta_n(t))\big) \end{aligned} \tag{6}$$

where $\beta_0 = (4\pi f d_0/c)$, $d_0 = 1$ m is the reference distance. P_T is the transmit power, and parameters k_{LoS} and k_{NLoS} are associated with η_ξ. We focus on the CU's downlink communication throughput. Furthermore, we adopt frequency division multiple access technology for multiple access communication. Traditional UAV communication networks often adopt a fly-hover-communicate protocol, which has been proven to have low-energy efficiency and suboptimal [35]. In contrast, the in-flight communication scheme can extend communication duration, thereby enhancing throughput. Consequently, considering the UAV provides communication in flight, the aggregated communication throughput can be formulated as follows:

$$\overline{R} = \int_0^T \sum_{n=1}^{N} \frac{B}{N}\log_2\left(1 + \frac{P_{R,n}(t)}{\sigma_0^2}\right)dt \tag{7}$$

where B denotes the channel bandwidth in Hz and σ_0^2 is the noise power at the receiver. The optimization goal of the communication is to maximize the communication throughput.

B. Sensing Model

For location estimation, $\mathbf{p}_k = [x_k, y_k, 0]^T$ needs to be determined through measurements of the time-delay taken for the echoes of at least three distinct distances [36]. This implies that for each sensing target, the UAV requires a minimum

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of three hovering points. In the sensing model, we consider L ($L \geq 3$) hovering points around each sensing target. The echo time-delay is denoted as $\tau_k(l)$, namely, the time it takes for the ISAC signals to be transmitted from the lth hovering point $[x_l, y_l, H]^T, l \in \{1, \ldots, L\}$ to the kth sensing target $[x_k, y_k, 0]^T, k \in \mathcal{K}$, and subsequently reflected back to the lth hovering point. We denote $d_k(l)$ as the distance from the UAV to the kth sensing target, which is

$$d_k(l) = \sqrt{H^2 + \|\left[x_l, y_l\right]^T - \left[x_k, y_k\right]^T\|^2}. \tag{8}$$

Thus, the distance vector from all L hovering points around the kth sensing targets is $\mathbf{d}_k = [d_k(l), \ldots, d_k(L)]^T$. The positional coordinates of the sensing target can be estimated based on the echo time-delay measurement mode [37]. The measured values of the echo time-delay can be represented as follows:

$$\hat{\tau}_k(l) = \tau_k(l) + w_k(l) \tag{9}$$

where $\tau_k(l)$ represents the actual echo time-delay and $w_k(l)$ denotes the Gaussian noise in measurement values, with a mean of zero and a variance of $\sigma_k^2(l)$. $\sigma_k^2(l)$ follows an independent Gaussian distribution, inversely proportional to the SNR of the ISAC signal echoes received [38]. The echo time-delay of all L hovering points of kth sensing target is composed of vector $\boldsymbol{\tau}_k = [\tau_k(1), \ldots, \tau_k(L)]^T$.

In radar sensing, in order to further derive the SNR at the receiver, it is necessary to consider the power gain of the bidirectional channel. Consequently, the total power of ISAC echoes reflected by the kth sensing target and received by the UAV at the lth hovering point is

$$P_{R,k} = \frac{P_T G^2 \lambda^2 G_p \sigma_{\mathrm{RCS}}}{4\pi^3 d_k(l)^4} \tag{10}$$

where G_p represents the gain introduced by signal processing (due to the receiving matched filter). λ represents the wavelength of the ISAC signal, G is the antenna gain, while σ_{RCS} refers to the radar cross section. Let $G_s = ([G^2\lambda^2 G_p \sigma_{\mathrm{RCS}}]/4\pi^3)$, the SNR can be further derived as

$$\mathrm{SNR}_k(l) = \frac{P_T G_s G_p}{\sigma_0^2 d_k(l)^4}. \tag{11}$$

$\sigma_k^2(l)$ is inversely proportional to $\mathrm{SNR}_k(l)$, and the scale coefficient is denoted as a_t. Then $\sigma_k^2(l)$ can be obtained as follows [37]:

$$\begin{aligned}\sigma_k^2(l) &= \frac{a_t}{8\pi^2 B^2} \frac{1}{\mathrm{SNR}_k(l)} \\ &= \frac{a_t \sigma_0^2 d_k(l)^4}{8\pi^2 B^2 P_T G_s G_p}.\end{aligned} \tag{12}$$

For the sensing tasks, the parameters to be estimated are the coordinates of the kth sensing target, which are represented as the vector $\mathbf{u}_k = [x_k, y_k]^T, k \in \mathcal{K}$. The CRB provides a lower bound for the MSE of any unbiased estimator for $\mathbf{u}_k$, while MSE often fails to display expression, making it difficult to minimize with optimization tools [39]. Therefore, we drive the CRB of the proposed system to assess the sensing performance. Given the vector parameters $\mathbf{u}_k$, the unbiased estimate $\hat{\mathbf{u}}_k$ satisfies the following inequality:

$$\mathbb{E}_{\mathbf{u}_k}\left\{\left(\hat{\mathbf{u}}_k - \mathbf{u}_k\right)\left(\hat{\mathbf{u}}_k - \mathbf{u}_k\right)^T\right\} \geq \mathbf{J}^{-1}(\mathbf{u}_k) \tag{13}$$

where $\mathbf{J}(\mathbf{u}_k)$ is the Fisher information matrix (FIM) of $\mathbf{u}_k$. Furthermore, it can be noted that

$$\mathbf{CRB}_{\mathbf{u}_k} = \mathbf{J}^{-1}(\mathbf{u}_k). \tag{14}$$

We can derive the relationship between the CRB of $\mathbf{u}_k$ and the sum of the CRB of x_k and y_k as follows [40]:

$$\mathrm{CRB}_{x_k} + \mathrm{CRB}_{y_k} = \mathrm{tr}\left(\mathbf{CRB}_{\mathbf{u}_k}\right). \tag{15}$$

Consequently, we can initially compute the $\mathbf{J}(\boldsymbol{\tau}_k)$ for $\boldsymbol{\tau}_k$ and then obtain $\mathbf{J}(\mathbf{u}_k)$ through the following chain rule:

$$\mathbf{J}(\mathbf{u}_k) = \mathbf{Q}\mathbf{J}(\boldsymbol{\tau}_k)\mathbf{Q}^T \tag{16}$$

where the Jacobian matrix $\mathbf{Q} = (\boldsymbol{\tau}_k^T/\partial \mathbf{u}_k)$, which can be expressed as

$$\mathbf{Q} = \begin{bmatrix} \frac{2(x_1 - x_k)}{c d_k(1)}, \ldots, \frac{2(x_L - x_k)}{c d_k(L)} \\ \frac{2(y_1 - y_k)}{c d_k(1)}, \ldots, \frac{2(y_L - y_k)}{c d_k(L)} \end{bmatrix}. \tag{17}$$

Meanwhile, the measurement vector $\hat{\boldsymbol{\tau}}_k$ follows a Gaussian distribution with a mean of $\boldsymbol{\tau}_k$ and a variance of $\mathbf{C}(\boldsymbol{\tau}_k)$. $\mathbf{C}(\boldsymbol{\tau}_k)$ is the measurement covariance as

$$\begin{aligned}\mathbf{C}(\boldsymbol{\tau}_k)) &= \frac{a_t}{8\pi^2 B^2} \mathrm{diag}\left(\frac{1}{\mathrm{SNR}_k(1)}, \ldots, \frac{1}{\mathrm{SNR}_k(L)}\right) \\ &= \frac{a_t \sigma_0^2}{8\pi^2 B^2 P_T G_S G_p} \mathrm{diag}\left(d_k(1)^4, \ldots, d_k(L)^4\right).\end{aligned} \tag{18}$$

Consequently, $\mathbf{J}(\boldsymbol{\tau}_k)$ can be determined through the following procedure [40]:

$$\begin{aligned}\mathbf{J}(\boldsymbol{\tau}_k) = &\left[\frac{\partial \boldsymbol{\tau}_k}{\partial \tau_k(p)}\right]^T \mathbf{C}^{-1}(\boldsymbol{\tau}_k)\left[\frac{\partial \boldsymbol{\tau}_k}{\partial \tau_k(q)}\right] \\ &+ \frac{1}{2}\mathrm{tr}\left[\mathbf{C}^{-1}(\boldsymbol{\tau}_k)\frac{\partial \mathbf{C}(\boldsymbol{\tau}_k)}{\partial \tau_k(p)}\mathbf{C}^{-1}(\boldsymbol{\tau}_k)\frac{\partial \mathbf{C}(\boldsymbol{\tau}_k)}{\partial \tau_k(q)}\right]\end{aligned} \tag{19}$$

where $p, q = 1, \ldots, L$. After simple mathematic operations, we can obtain $\mathbf{CRB}_{\mathbf{u}_k}$

$$\mathbf{CRB}_{\mathbf{u}_k} = \mathbf{J}^{-1}(\mathbf{u}_k) = \frac{g_a + g_b}{g_a g_b - g_c^2} \tag{20}$$

where the elements g_a, g_b, and g_c are defined as

$$g_a = \sum_{l=1}^{L}\left\{\frac{\alpha}{d_k(l)^6}(x_l - x_k)^2 + \frac{8}{d_k(l)^4}(x_l - x_k)^2\right\} \tag{21}$$

$$g_b = \sum_{l=1}^{L}\left\{\frac{\alpha}{d_k(l)^6}(y_l - y_k)^2 + \frac{8}{d_k(l)^4}(y_l - y_k)^2\right\} \tag{22}$$

$$g_c = \sum_{l=1}^{L}\left\{\frac{\alpha}{d_k(l)^6}(x_l - x_k)(y_l - y_k) + \frac{8}{d_k(l)^4}(x_l - x_k)(y_l - y_k)\right\} \tag{23}$$

where $\alpha = ([32\pi^2 B^2 P_T G_S G_p]/\sigma_0^2 c^2 a_t)$.

In conclusion, in the considered ISAC system, the objective of sensing optimization is to minimize the average CRB for all sensing targets' coordinates estimation errors, thereby optimizing the overall sensing performance of the system.

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We employ the maximum likelihood estimation (MLE) method to estimate the sensing targets' coordinates. The measured values of the echo time-delay follow the Gaussian distribution and are statistically independent of each other. When sensing the kth sensing target, the echo time-delay measurement vector $\hat{\boldsymbol{\tau}}_k$ is composed of L distinct echo time-delay measurements, $\hat{\boldsymbol{\tau}}_k = [\hat{\tau}_k(1), \ldots, \hat{\tau}_k(L)]^T$. Thus, a joint likelihood function for $\hat{\boldsymbol{\tau}}_k$ can be established, and the parameters $[\hat{x}_k, \hat{y}_k]^T$ that maximize the likelihood function represent the MLE for the sensing target. Hence, a joint likelihood function can be formulated as

$$p(\hat{\boldsymbol{\tau}}_k|[x_k, y_k]^T) = \frac{1}{(2\pi\sigma_k(l)^2)^{\frac{L}{2}}} \exp\left(-\frac{1}{2\sigma_k(l)^2}\sum_{l=1}^{L}\left(\hat{\tau}_k(l) - \frac{2\|\mathbf{q}_l - \mathbf{p}_k\|}{c}\right)^2\right). \quad (24)$$

Therefore, the estimated coordinates of the kth sensing target can be obtained as

$$\left[\hat{x}_k, \hat{y}_k\right]^T = \arg\left\{\max \ln p\left(\hat{\boldsymbol{\tau}}_k|\left[x_k, y_k\right]^T\right)\right\}. \quad (25)$$

To obtain the above estimation, it is necessary to fulfill

$$\frac{\partial}{\partial\left[x_k, y_k\right]^T} \ln p\left(\hat{\boldsymbol{\tau}}_k|\left[x_k, y_k\right]^T\right) = 0|_{[x_k, y_k]^T = [\hat{x}_k, \hat{y}_k]^T}. \quad (26)$$

Additionally, when $[x_k, y_k]^T = [\hat{x}_k, \hat{y}_k]^T$, it is required that the second derivative of $\ln p(\hat{\boldsymbol{\tau}}_k|[x_k, y_k]^T)$ is less than 0. The above expression is challenging to obtain a closed-form analytical solution, therefore, numerical methods or the EM algorithm can be employed. Additionally, the estimation performance of MLE can be evaluated through Monte Carlo simulations.

C. UAV Propulsion Power Model

In the ISAC system, due to the limitations of the physical size of the UAV, the onboard energy of the UAV must be optimized for task execution. The overall energy consumption of the UAV typically consists of three primary components: 1) flight; 2) hovering; and 3) signals transmission. The power consumption related to the propulsion of the UAV can be modeled as a function of the UAV's velocity V [5]

$$P(V) = P_0\left(1 + \frac{3V^2}{U_{\text{tip}}^2}\right) + P_1\left(\sqrt{1 + \frac{V^4}{4v_0^4}} - \frac{V^2}{2v_0^2}\right)^{\frac{1}{2}} + \frac{1}{2}D_0\rho s A V^3 \quad (27)$$

where P_0 and P_1 represent the blade profile power and induced power in hovering status, respectively. U_{tip} denotes the rotor tip velocity. v_0 stands for the average rotor induced velocity in hover. D_0 and s are the fuselage drag ratio and rotor solidity, respectively. And ρ and A are the air density and rotor disc area, respectively. Furthermore, the power consumption during the hovering of the UAV corresponds to the scenario where the velocity $V = 0$ in the above equation

$$P(0) = P_0 + P_1. \quad (28)$$

Additionally, the energy consumption associated with communication is determined by the signal transmission power. Typical values of the parameters in (27) are listed in Table I [5].

TABLE I
TYPICAL VALUES OF THE PARAMETERS IN (27)

parameter	value	parameter	value
P_0	80W	D_0	0.6
U_{tip}	120m/s	ρ	1.225kg/m^3
P_1	88.6W	s	0.05m^3
v_0	4.03m/s	A	0.503m^2

D. ISAC Path Discretization Framework

In the considered ISAC system, both communication and sensing performance rely on trajectory $\mathbf{q}(t)$ of the UAV. Therefore, we initially address the framework of the UAV trajectory. Within the operational time T, $\mathbf{q}(t)$ essentially represents an infinite set of continuous waypoints for the UAV. To this end, prior works mostly adopted the method of time discretization, where the time horizon $[0, T]$ is discretized into a finite number of time slots with sufficiently small slot length [26], [41]. However, we employ a path discretization framework to address the UAV trajectory problem in our approach. This entails dividing the UAV trajectory into $(M + 1)$ segments, thereby transforming the continuous optimization variables $\mathbf{q}(t)$ associated with the UAV trajectory into $(M + 2)$ discrete optimization variables $\mathbf{q}_m$, where $m \in \{1, \ldots, M + 2\}$. $\mathbf{q}$ represents the set of $M + 2$ waypoints for the UAV, with $\mathbf{q}_1 = \mathbf{q}_I$ and $\mathbf{q}_{M+2} = \mathbf{q}_F$. In contrast to time discretization schemes, path discretization schemes adaptively determine time intervals rather than imposing fixed time intervals. Furthermore, path discretization can reduce the number of optimization variables, especially when the UAV needs to hover or fly at low speeds.

Following this, we restructure the previously described system performance functions using the path discretization framework. Denoting the distance between the UAV and the nth CU in the mth segment as $d_{m,n} = \|\mathbf{q}_m - \mathbf{w}_n\|^2, m \in \{1, \ldots, M + 1\}, n \in \mathcal{N}$, assuming that the position of the UAV within a segment remains consistent with the segment's starting point, the communication rate can be reformulated as

$$R_{m,n} = \frac{B}{N}\log_2\left(1 + \frac{P_T d_{m,n}^{-2}\beta_0^{-2}\sum_\xi k_\xi P(\xi, \theta_{m,n})}{\sigma_0^2}\right). \quad (29)$$

We define $\gamma_0 = ([P_T\beta_0^{-2}\sum_\xi k_\xi P(\xi, \theta_{m,n}))]/\sigma_0^2)$, thus, upon rearrangement

$$R_{m,n} = \frac{B}{N}\log_2\left(1 + \frac{\gamma_0}{\|\mathbf{q}_m - \mathbf{w}_n\|^2}\right). \quad (30)$$

Let t_m denote the time the UAV stays in the mth segment. With this, we can calculate the total communication throughput as follows:

$$\overline{R} = \sum_{m=0}^{M+1}\sum_{n=1}^{N} t_m R_{m,n}. \quad (31)$$

Similarly, assuming the UAV maintains a constant velocity within each segment of the path discretization and preserves the velocity at the starting point of each segment, this assumption is reasonable due to the short length of each segment. Under this assumption, denote the velocity of the UAV within the mth segment as v_m. Therefore, the relationship between

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velocity and t_m within the mth segment is given by $v_m = ([\|\mathbf{q}_{m+1} - \mathbf{q}_m\|]/t_m) \quad \forall m$. The energy power consumption of the UAV flight in the mth segment can be obtained as follows:

$$P(v_m) = P_0\left(1 + \frac{3v_m^2}{U_{\text{tip}}^2}\right) + P_1\left(\sqrt{1 + \frac{v_m^4}{4v_0^4}} - \frac{v_m^2}{2v_0^2}\right)^{\frac{1}{2}} + \frac{1}{2}D_0\rho sAv_m^3. \tag{32}$$

III. Problem Formulation

In this section, we begin by organizing the functionalities that need to be implemented within the described ISAC system, along with the resource constraints. In prior research, ISAC systems generally have three different typical application scenarios: 1) communication centric; 2) sensing centric; and 3) tradeoff of C&S [8]. We will thus establish three corresponding types of ISAC optimization problems and then propose a generic solution in Section IV.

A. Communication-Centric Scenarios

In the communication-centric scenarios, the primary requirement is to maximize the communication throughput, while meeting the sensing metrics. Therefore, we formulate the following objective function:

$$\max_{\mathbf{q},\mathbf{t}} \overline{R} \tag{33}$$

where $\mathbf{t}$ is a vector of flight time in each segment and $\mathbf{t} = [t_1, \ldots, t_{M+1}]^T$. The overall sensing performance constraint is defined to ensure that the average CRB for all sensing targets meets a specified threshold $\eta_{\max}$

$$\frac{\sum_{k=1}^{K} \text{CRB}_{x_k,y_k}}{K} \le \eta_{\max}. \tag{34}$$

Assuming that the battery capacity of the UAV is $E_{\max}$, the UAV energy consumption in the whole process should meet the following constraint:

$$\sum_{m=0}^{M+1} t_m P(v_m) + \text{LKT}_h P(0) + P_T T \le E_{\max} \tag{35}$$

where T_h is the hovering time of the UAV at each hovering waypoint. The first item on the left is the energy consumed by UAV flight denoted by E_f, the second part is the energy consumed by hovering for targets' localization, denoted by E_h, and the third part is the energy consumed by transmitting ISAC signals denoted by E_c. In summary, in the communication-centric ISAC scenarios, the optimization problem that needs to be solved is as follows:

$$\text{P1}: \max_{\mathbf{q},\mathbf{t}} \ \overline{R} \tag{36a}$$

s.t.

$$E_f + E_h + E_c \le E_{\max} \tag{36b}$$

$$\|\mathbf{q}_{m+1} - \mathbf{q}_m\| \le \min\{\Delta_{\max}, T_m V_{\max}\} \quad \forall m \tag{36c}$$

$$\mathbf{q}_1 = \mathbf{q}_I, \mathbf{q}_{M+2} = \mathbf{q}_F \tag{36d}$$

$$\sum_{m=1}^{M+1} t_m + \text{LKT}_h \le T \tag{36e}$$

$$t_m \ge 0 \quad \forall m$$

(34). (36f)

In accordance with (36c), the distance between two waypoints is kept short enough that it allows us to assume that the UAV flies at a constant speed v_m and relative position $\mathbf{q}_m$ with respect to the sensing target or CU remain unchanged between these two waypoints. $\Delta_{\max}$ satisfies $\Delta_{\max} \ll H$. $V_{\max}$ is the maximum speed of the UAV. Constraints (36e) and (36f) enable the UAV to complete the ISAC tasks within the mission time.

B. Sensing-Centric Scenarios

In the sensing-centric scenarios, the primary focus is to achieve high-precision localization while ensuring that the communication throughput exceeds a tolerable minimum. Therefore, the objective is to minimize the average CRB, subject to the constraint of achieving communication throughput greater than the minimum $Q_{\min}$. Finally, similar to the previous section, constraints related to UAV motion are incorporated. In summary, the problem model for the sensing-centric scenarios can be formulated as follows:

$$\text{P2}: \min_{\mathbf{q},\mathbf{t}} \ \frac{\sum_{k=1}^{K} \text{CRB}_{x_k,y_k}}{K} \tag{37a}$$

s.t.

$$\sum_{m=0}^{M+1} t_m R_{m,n} \ge Q_{\min}$$

(36b), (36c), (36d), (36e), (36f) (37b)

Here, the difference between communication-centric and sensing-centric scenarios is clearly evident, achieved by interchanging the objective function and constraints, thus shifting the system's primary focus from communication to sensing. It is noteworthy that although both scenarios emphasize either communication or sensing as their primary function, they both need to meet the system requirements for another functionality.

C. C&S Tradeoff Scenarios

In terms of C&S performance, the first two application scenarios can be regarded as allocating more resources to prioritize one of the functions, while meeting the minimum requirements for the other. In this perspective, these application scenarios are essentially designed for the specialized ISAC applications. However, if we consider a more profound integration of C&S, how should ISAC systems be designed? This leads to the currently popular C&C tradeoff application scenarios, where the scenario does not predefine minimum standards for C&S but allocates all system resources to coordinate them, aiming to achieve an optimal tradeoff between C&S.

When constructing the problem model, the specific approach involves normalizing $\overline{R}$ and $([\sum_{k=1}^{K} \text{CRB}_{x_k,y_k}]/K)$. Then, these normalized metrics are normalized and adjusted by w_s and w_c and used as the objective function, in addition to the constraints related to UAV motion. With these considerations,

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we can formulate the problem model for the tradeoff scenarios as follows:

$$\text{P3}: \min_{\mathbf{q},\mathbf{t}} \ w_s \frac{\sum_{k=1}^{K} \text{CRB}_{x_k,y_k}}{K\widetilde{\text{CRB}}_{x_k,y_k}} - w_c \frac{\overline{R}}{\widetilde{R}}$$
$$\text{s.t.}$$
$$(36b), (36c), (36d), (36e), (36f) \tag{38}$$

where $\widetilde{\text{CRB}}_{x_k,y_k}$ and $\widetilde{R}$ are normalization parameters associated with communication and sensing performance metrics, respectively. Note that $w_s + w_c = 1$, w_c and w_s, respectively, indicate the resource allocation preference of the ISAC application scenario toward C&S, determined based on application requirements.

D. Optimization of the Maximum CRB for All Targets

The average CRB can represent the average sensing target localization accuracy of the network, reflecting the overall sensing performance. Furthermore, we are also concerned with the worst-case of each sensing target's localization accuracy, which involves a fairness optimization problem. We aim to enhance the worst-case of the localization accuracy for each sensing target by minimizing the maximum CRB for all sensing targets. Specifically, we introduce an auxiliary variable ξ to represent the maximum of the CRB for all sensing targets. Under the framework of optimizing the maximum CRB for all targets, P1, P2, and P3 can be, respectively, rewritten as

$$\text{P4}: \max_{\mathbf{q},\mathbf{t}} \ \overline{R} \tag{39a}$$
$$\text{s.t.}$$
$$\xi \le \eta_{\max} \tag{39b}$$
$$\text{CRB}_{x_k,y_k} \le \xi \quad \forall k$$
$$(36b), (36c), (36d), (36e), (36f) \tag{39c}$$

$$\text{P5}: \min_{\mathbf{q},\mathbf{t}} \ \xi$$
$$\text{s.t.}$$
$$(36b), (36c), (36d), (36e), (36f), (37b), (39c) \tag{40}$$

$$\text{P6}: \min_{\mathbf{q},\mathbf{t}} \ w_s \frac{\xi}{\widetilde{\text{CRB}}_{x_k,y_k}} - w_c \frac{\overline{R}}{\widetilde{R}}$$
$$\text{s.t.}$$
$$(36b), (36c), (36d), (36e), (36f), (39c). \tag{41}$$

IV. Proposed Solutions

In this section, a generic solution is proposed for the three UAV-enabled ISAC scenarios, which are elaborated in Section III. First, an initialization scheme for the trajectory based on the TSP is introduced to generate the association between the UAV and sensing targets or CUs and the initial trajectory. Subsequently, given the established nonconvex optimization problems, an iterative algorithm based on SCA is employed to obtain the optimal approximation. Finally, a UAV trajectory replanning scheme is developed to further enhance the performance.

A. TSP-Based Trajectory Initialization

From the derived communication and sensing metric presented earlier, it is evident that the system's performance is closely related to the trajectory of the UAV. More specifically, shorter distance between the UAV and CUs or sensing targets will bring better performance. Furthermore, the trajectory of the UAV is subject to energy constraint. This implies that shorter flight distance for the UAV is more likely to satisfy the energy constraint. Based on the above analysis, considering the scenario that sequentially approaches each CU and sensing target while minimizes the flight distance can be viewed as a typical variant of TSP problem [42]. Therefore, we propose a trajectory initialization scheme that obtains the initial trajectory of the UAV by solving a modified TSP problem.

First, we determine the positions that need to be traversed in the modified TSP problem. The first component consists of the set of location points $\mathbf{Q}_c$ corresponding to CUs. The second component comprises the positions of the sensing targets, which are unknown but can be initialized based on historical data. Subsequently, we randomly select L distinct location points within the surrounding region S_k of each initial estimated sensing target position. These selected points will serve as the set of locations $\mathbf{Q}_t$. Finally, including the starting point $\mathbf{q}_I$ and the end point $\mathbf{q}_F$, we compile the set of positions to be traversed in the modified TSP problem as $\mathbf{Q}_{tsp} = \{\mathbf{q}_I, \mathbf{q}_F, \mathbf{Q}_c, \mathbf{Q}_t\}$, the number of location points is given by $N_{tsp} = LK + N + 2$. In that case, we will obtain the initial user access sequence by solving the following problem:

$$\text{P7}: \min_{\boldsymbol{\pi},\boldsymbol{\varsigma}} \ \sum_{i=1}^{N_{tsp}} \sum_{i \ne j} d_{ij}\pi_{ij} \tag{42a}$$
$$\text{s.t.}$$
$$\pi_{ii} = 0 \tag{42b}$$
$$\sum_{i=1}^{N_{tsp}} \pi_{ij} = 1 \ \ \forall i \in \{1, \ldots, N_{tsp}\} \tag{42c}$$
$$\sum_{j=1}^{N_{tsp}} \pi_{ij} = 1 \ \ \forall j \in \{1, \ldots, N_{tsp}\} \tag{42d}$$
$$\varsigma_i - \varsigma_j + N_{tsp}\pi_{ij} \le N_{tsp} - 1 \tag{42e}$$
$$\varsigma_1 = 1, \varsigma_{N_{tsp}} = N_{tsp} \tag{42f}$$
$$2 \le \varsigma_i \le \left(N_{tsp} - 1\right) \ \ \forall i \in \left\{2, \ldots, \left(N_{tsp} - 1\right)\right\} \tag{42g}$$

where d_{ij} represents the distance between location points i and j. π_{ij} is a binary variable representing whether there is a path from location point i to location point j, with a value of 1 indicating the existence of a path and 0 otherwise. Moreover, ς_i is used to represent the order of location point i. In the case of the modified TSP problem, it can be addressed through adaptations of classical algorithms [42].

Next, let us summarize the advantages of this approach. First, the initial UAV trajectories obtained using the aforementioned initialization method lie within the vicinity of the optimal solutions for P1, P2, P3, P4, P5 and P6, which significantly enhances the speed of problem-solving. Second, from the obtained visiting order of location points, we can

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extract the association matrix between the UAV and the CUs or sensing targets, avoiding the need to address integer programming problems. This significantly reduces the complexity of problem-solving. Lastly, the elevation angle variables $\theta_{m,n}$ can be replaced by the average elevation angle $\bar{\theta}$ using heuristic algorithms, consequently obtaining the average probability $P_{LoS}(\bar{\theta})$ of LoS links.

B. Proposed Algorithm for UAV-ISAC

The objectives and constraints involved in problems P1, P2, and P3, which include CRB_{x_k,y_k}, $\overline{R}$, and E_f, are nonconvex and challenging to solve. We propose a unified iterative algorithm to obtain their best approximate solutions. The main idea is to utilize local convex approximation to handle nonconvex portions and then employ a SCA technique for optimization [43].

First, the total communication throughput $\overline{R} = \sum_{m=0}^{M+1}\sum_{n=1}^{N} t_m R_{m,n}$, which is nonconvex but a concave function, can be approximated using its first-order Taylor expansion. We introduce an auxiliary variable $A_{m,n}$

$$A_{m,n}^2 = t_m R_{m,n} = t_m \frac{B}{N}\log_2\left(1+\frac{\gamma_0}{\|\mathbf{q}_m-\mathbf{w}_n\|^2}\right). \tag{43}$$

$A_{m,n}^2$ is a convex function with respect to $A_{m,n}$, and its lower bound can be obtained through the first-order Taylor expansion. In the rth iteration of the SCA, it can be expressed as

$$A_{m,n}^2 \geq A_{m,n}^{(r)2} + 2A_{m,n}^{(r)}\left(A_{m,n}-A_{m,n}^{(r)}\right). \tag{44}$$

Therefore, $\sum_{m=0}^{M+1}\sum_{n=1}^{N} t_m R_{m,n}$ can be replaced by its lower bound. Meanwhile, we need to add the following constraint:

$$\frac{A_{m,n}^2}{t_m}\frac{N}{B} \leq \log_2\left(1+\frac{\gamma_0}{\|\mathbf{q}_m-\mathbf{w}_n\|^2}\right) \tag{45}$$

where $\log_2(1+\gamma_0/[\|\mathbf{q}_m-\mathbf{w}_n\|^2])$ can be lower bounded through first-order Taylor expansion, and at the rth iteration

$$\begin{aligned}
&\log_2\left(1+\frac{\gamma_0}{\|\mathbf{q}_m-\mathbf{w}_n\|^2}\right)\\
&\geq \log_2\left(1+\frac{\gamma_0}{\|\mathbf{q}_m{}^{(r)}-\mathbf{w}_n\|^2}\right)\\
&\quad-\beta_{m,n}\Big(\|\left[x_m^{(r)},y_m^{(r)}\right]^T-\left[x_n,y_n\right]^T\|^2\\
&\quad-\|\left[x_m,y_m\right]^T-\left[x_n,y_n\right]^T\|^2\Big)
\end{aligned} \tag{46}$$

$$\beta_{m,n} = \frac{\log_2(e)\gamma_0}{\|\mathbf{q}_m{}^{(r)}-\mathbf{w}_n\|^2\left(\|\mathbf{q}_m{}^{(r)}-\mathbf{w}_n\|^2+\gamma_0\right)}. \tag{47}$$

Designate the right-hand side of the aforementioned inequality as $U_L{}^{(r)}$. Subsequently, we address the nonconvex average CRB expression. As the expression involves K as a constant, we proceed by directly performing a first-order Taylor expansion of CRB_{x_k,y_k}, setting $\mathrm{CRB}_{x_k,y_k} = f(\mathbf{x},\mathbf{y})$. In the rth iteration, we have

$$\begin{aligned}
\mathrm{CRB}_{x_k,y_k} &\approx f\left(\mathbf{x}^{(r)},\mathbf{y}^{(r)}\right)+\sum_{l=1}^{L}\nabla f(\mathbf{x},\mathbf{y})_{x_l}\left(x_l-x_l^{(r)}\right)\\
&\quad+\sum_{l=1}^{L}\nabla f(\mathbf{x},\mathbf{y})_{y_l}\left(y_l-y_l^{(r)}\right)=\hat{\mathrm{CRB}}_{x_k,y_k}.
\end{aligned} \tag{48}$$

To meet the requirements for both approximation and convergence, it is essential to set the update stepsize $\triangle r^{(r)}$ ($\triangle r^{(r)} \in [0,1]$) for $\mathbf{x}^{(r)}$ and $\mathbf{y}^{(r)}$ [44]. Consequently, we introduce a cosine step size, resembling the Cosine Annealing learning rate [45]. Finally, the energy consumption constraint of the UAV, where only the leftmost term is nonconvex, can be expanded as follows:

$$\begin{aligned}
\sum_{m=0}^{M+1} t_m P(v_m) &= \sum_{m=0}^{M+1}\Bigg\{t_m P_0\left(1+\frac{3v_m^2}{U_{\mathrm{tip}}^2}\right)\\
&\quad+t_m P_1\left(\sqrt{1+\frac{v_m^4}{4v_0^4}}-\frac{v_m^2}{2v_0^2}\right)^{\frac{1}{2}}+\frac{1}{2}D_0\rho s A v_m^3 t_m\Bigg\}.
\end{aligned} \tag{49}$$

Due to the condition $v_m = ([\|\mathbf{q}_{m+1}-\mathbf{q}_m\|]/t_m)$, we can deduce $v_m t_m = \triangle m$. By substitute this into the previous equation, we can rewrite it as follows:

$$\begin{aligned}
\sum_{m=0}^{M+1} t_m P(v_m) &= \sum_{m=0}^{M+1}\Bigg\{P_0\left(t_m+\frac{3\triangle m^2}{U_{\mathrm{tip}}^2 t_m}\right)\\
&\quad+P_1\left(\sqrt{t_m^4+\frac{\triangle m^4}{4v_0^4}}-\frac{\triangle m^2}{2v_0^2}\right)^{\frac{1}{2}}+\frac{1}{2}D_0\rho s A\frac{\triangle m^3}{t_m{}^2}\Bigg\}.
\end{aligned} \tag{50}$$

The second term in the expression is the only nonconvex component and requires additional operations. Here, we introduce auxiliary variables Z_m

$$Z_m = \left(\sqrt{t_m^4+\frac{\triangle m^4}{2v_0^2}}-\frac{\triangle m^2}{2v_0^2}\right)^{\frac{1}{2}}, Z_m \geq 0. \tag{51}$$

So (50) can be written as

$$\begin{aligned}
\sum_{m=0}^{M+1} t_m P(v_m) &= \sum_{m=0}^{M+1}\Bigg\{P_0\left(t_m+\frac{3\triangle m^2}{U_{\mathrm{tip}}^2 t_m}\right)\\
&\quad+P_1 Z_m+\frac{1}{2}D_0\rho s A\frac{\triangle m^3}{t_m{}^2}\Bigg\}=\hat{E}_f.
\end{aligned} \tag{52}$$

Due to $Z_m^2 = \sqrt{t_m^4+(\triangle m^4/2v_0^2)}-(\triangle m^2/2v_0^2)$, we obtain $(t_m^4/Z_m^2) = Z_m^2+(\triangle m^2/v_0^2)$, thus introducing the constraint

$$\frac{t_m^4}{Z_m^2} \leq Z_m^2+\frac{\triangle m^2}{v_0^2} = Z_m^2+\frac{\|\mathbf{q}_{m+1}-\mathbf{q}_m\|^2}{v_0^2}. \tag{53}$$

Subsequently, a first-order Taylor expansion is performed on $\|\mathbf{q}_{m+1}-\mathbf{q}_m\|^2$ and Z_m^2 to obtain its lower

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bound, which can be described as follows in the rth iteration:

$$\begin{aligned} & Z_m^2 + \frac{\|\mathbf{q}_{m+1} - \mathbf{q}_m\|^2}{v_0^2} \\ & \geq Z_m^{(r)2} + \frac{\|\mathbf{q}_{m+1}^{(r)} - \mathbf{q}_m^{(r)}\|^2}{v_0^2} \\ & \quad + 2\frac{\|\mathbf{q}_{m+1}^{(r)} - \mathbf{q}_m^{(r)}\|^2}{v_0^2}\left(\|\mathbf{q}_{m+1} - \mathbf{q}_m\| - \|\mathbf{q}_{m+1}^{(r)} - \mathbf{q}_m^{(r)}\|\right) \\ & \quad + 2Z_m^{(r)}\left(Z_m - Z_m^{(r)}\right). \end{aligned} \tag{54}$$

Denote the right-hand side of the above inequality as $O_m{}^{(r)}$, and based on the above analysis, the optimization problems P1, P2, and P3 can be transformed into the following problems to be addressed using SCA:

$$\text{P1.1}: \max_{\mathbf{q},\mathbf{t},\mathbf{A},\mathbf{Z}} \sum_{m=0}^{M+1}\sum_{n=1}^{N} A_{m,n}^{(r)2} + 2A_{m,n}^{(r)}\left(A_{m,n} - A_{m,n}^{(r)}\right) \tag{55a}$$

s.t.

$$\frac{A_{m,n}^2}{t_m}\frac{N}{B} \leq U_L{}^{(r)} \tag{55b}$$

$$\frac{\sum_{k=1}^{K} \hat{\text{CRB}}_{x_k,y_k}}{K} \leq \eta_{\max} \tag{55c}$$

$$\hat{E}_f + E_h + E_c \leq E_{\max} \tag{55d}$$

$$\frac{t_m^4}{Z_m^2} \leq O_m{}^{(r)} \tag{55e}$$

$$\|\mathbf{q}_{m+1} - \mathbf{q}_m\| \leq \min\{\Delta_{\max}, t_m V_{\max}\} \quad \forall m \tag{55f}$$

$$\mathbf{q}_1 = \mathbf{q}_I, \mathbf{q}_{M+2} = \mathbf{q}_F \tag{55g}$$

$$\sum_{m=1}^{M+1} t_m + \text{LKT}_h \leq T \tag{55h}$$

$$t_m \geq 0 \quad \forall m. \tag{55i}$$

$$\text{P2.1}: \min_{\mathbf{q},\mathbf{t},\mathbf{A},\mathbf{Z}} \frac{\sum_{k=1}^{K} \hat{\text{CRB}}_{x_k,y_k}}{K} \tag{56a}$$

s.t.

$$\sum_{m=0}^{M+1} A_{m,n}^{(r)2} + 2A_{m,n}^{(r)}\left(A_{m,n} - A_{m,n}^{(r)}\right) \geq Q_{\min}$$

(55b), (55d), (55e)

(55f), (55g), (55h), (55i). \tag{56b}

$$\text{P3.1}: \min_{\mathbf{q},\mathbf{t},\mathbf{A},\mathbf{Z}} w_s \frac{\sum_{k=1}^{K} \hat{\text{CRB}}_{x_k,y_k}}{K\widetilde{\text{CRB}}_{x_k,y_k}} - w_c \frac{\sum_{m=0}^{M+1}\sum_{n=1}^{N} A_{m,n}^{(r)2} + 2A_{m,n}^{(r)}\left(A_{m,n} - A_{m,n}^{(r)}\right)}{\widetilde{R}}$$

s.t.

(55b), (55d), (55e)

(55f), (55g), (55h), (55i). \tag{57a}

It can be verified that P1.1, P2.1 and P3.1 are convex optimization problems, so they can be solved efficiently using standard convex optimization techniques or existing software toolkits, such as CVX. By solving P1.1, P2.1 and P3.1 and updating the local points successively in each iteration, an efficient algorithm for the original problem P1, P2 and P3 is obtained.

For the framework of optimizing the maximum CRB, replace the $(\sum_{k=1}^{K} \hat{CRB}_{x_k,y_k}/K)$ in P1.1, P2.1, and P3.1 with the maximum CRB of all sensing targets represented by auxiliary variable ξ, and add the following constraint:

$$\hat{\text{CRB}}_{x_k,y_k} \leq \xi. \tag{58}$$

Therefore, the UAV trajectory design optimization problem in the three typical ISAC scenarios can be formulated as follows:

$$\text{P4.1}: \max_{\mathbf{q},\mathbf{t},\mathbf{A},\mathbf{Z}} \sum_{m=0}^{M+1}\sum_{n=1}^{N} A_{m,n}^{(r)2} + 2A_{m,n}^{(r)}\left(A_{m,n} - A_{m,n}^{(r)}\right)$$

s.t.

(39b), (55b), (55d), (55e)

(55f), (55g), (55h), (55i), (58) \tag{59}

$$\text{P5.1}: \min_{\mathbf{q},\mathbf{t},\mathbf{A},\mathbf{Z}} \xi$$

s.t.

(55b), (55d), (55e), (55f)

(55g), (55h), (55i), (56b), (58) \tag{60}

$$\text{P6.1}: \min_{\mathbf{q},\mathbf{t},\mathbf{A},\mathbf{Z}} w_s \frac{\xi}{\widetilde{\text{CRB}}_{x_k,y_k}} - w_c \frac{\overline{R}}{\widetilde{R}}$$

s.t.

(55b), (55d), (55e), (55f)

(55g), (55h), (55i), (58). \tag{61}

Since (58) and (39b) are linear constraints, P4.1, P5.1, and P6.1 are convex problems and can be solved using standard convex optimization tools.

C. Trajectory Replanning

The initial information obtained by the system is usually limited, and if the UAV can assist in real-time trajectory planning based on the information they obtain, the system will achieve enhanced performance. Henceforth, we utilize the position information of each sensing target sensed during hovering to replan the subsequent trajectory of the UAV.

In our framework, the primary concept underlying trajectory replanning is segmented planning. In the initial UAV trajectory planning, we utilize the initial sensing target estimations denoted as $\hat{\mathbf{p}}_k^0$ to solve optimization problems, yielding the trajectory for the first planning. Subsequently, the first segment of the UAV trajectory is extracted from the first planning, encompassing the starting point to the $\lceil (L/2) \rceil$th hovering point (half of all hovering points) around the first sensing target. After the UAV completes the first segment, the estimation for the first sensing target is updated to $\hat{\mathbf{p}}_1^1$ based on information from the first $\lceil (L/2) \rceil$ hovering points. Subtracting the resources consumed during the first segment (e.g., energy, time), the updated estimation is then substituted back into

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Algorithm 1 Proposed Overall Algorithm for UAV-Enabled ISAC Networks

Initialize: $\hat{\mathbf{p}}_k^0$, the initial trajectory $\mathbf{q}_0$ and $\bar{\theta}$ from solving P7, w_c, w_s, η_{max}, Q_{min}, the tolerance ε, maximum iteration number r_{max}.

repeat

1) The UAV starts from the start point $\mathbf{q}_I$
2) for $i = 1 : K$
3) **repeat**
4) Given $\mathbf{q}^r$, $\mathbf{A}^r$, $\mathbf{t}^r$ and $\mathbf{Z}^r$, solve the UAV trajectory design problem and denote the optimal solution $\mathbf{q}^{r+1}$, $\mathbf{A}^{r+1}$, $\mathbf{t}^{r+1}$ and $\mathbf{Z}^{r+1}$. Update $r = r + 1$.

until The fractional increase of the objective value is below ε or $r \geq r_{max}$.

5) $\mathbf{q}_i = \mathbf{q}_I \rightarrow \mathbf{q}_{i,\lceil \frac{L}{2} \rceil}$
6) Update coordinates estimation $\hat{\mathbf{p}}_k^i$ based on $\boldsymbol{\tau}_i$ via MLE.
7) $i = i + 1$
8) Calculate E_{max}^i, T^i
9) $\mathbf{q}_I = \mathbf{q}_{i,\lceil \frac{L}{2} \rceil}$
10) **until** $(i > K)$
11) $\mathbf{q} = \sum_{i=1}^{K} \mathbf{q}_i$

optimization problems to obtain the UAV trajectory starting from the $(L/2)$th hovering point of the first sensing target to the endpoint. The second segment of the trajectory is extracted from this, encompassing the $(L/2)$th hovering point of the first sensing target to the $(L/2)$th hovering point of the second sensing target. This process is repeated iteratively until the UAV reaches the endpoint. Note that it is essential to ensure $L \geq 6$.

The benefits of this approach are evident. The UAV updates the estimations of each sensing target's coordinates based on the first $\lceil (L/2) \rceil$ points. Subsequently, the UAV's trajectory is replanned using these updated coordinates estimations, resulting in an updated UAV trajectory. By utilizing information updated from the first $\lceil (L/2) \rceil$ hovering points for each sensing target to replan the subsequent hovering points, the accuracy of sensing target coordinates estimations is enhanced. Denoting the intermediate hovering point of the kth sensing target as $\mathbf{q}_{k,\lceil (L/2) \rceil}$ the proposed algorithm can be summarized as Algorithm 1.

D. Overall Algorithm and Computational Complexity

Based on the above discussion, we propose an iterative algorithm for UAV trajectory design in the three typical ISAC scenarios, detailed in Algorithm 1. This algorithm directly solves problems P1.1, P2.1, P3.1, P4.1, P5.1, and P6.1, which approximate the original problems. Thus, the solutions from Algorithm 1 are approximate optimal solutions to the original problems, requiring an analysis of convergence. Denote $S(\mathbf{q}^r, \mathbf{A}^r, \mathbf{t}^r, \mathbf{Z}^r)$ be the objective value in the rth iteration and $S(\mathbf{q}^{r+1}, \mathbf{A}^{r+1}, \mathbf{t}^{r+1}, \mathbf{Z}^{r+1})$ in the $(r+1)$th iteration. By inputting the solution $\mathbf{q}^r, \mathbf{A}^r, \mathbf{t}^r, \mathbf{Z}^r$ of the rth iteration, the objective value is maximized in the communication-centric scenario, hence $S(\mathbf{q}^{r+1}, \mathbf{A}^{r+1}, \mathbf{t}^{r+1}, \mathbf{Z}^{r+1}) \geq S(\mathbf{q}^r, \mathbf{A}^r, \mathbf{t}^r, \mathbf{Z}^r)$. In the sensing-centric and tradeoff scenarios, the objective value is minimized, hence $S(\mathbf{q}^{r+1}, \mathbf{A}^{r+1}, \mathbf{t}^{r+1}, \mathbf{Z}^{r+1}) \leq S(\mathbf{q}^r, \mathbf{A}^r, \mathbf{t}^r, \mathbf{Z}^r)$. Therefore, the objective value in the communication-centric scenario is nondecreasing with a finite upper bound, and in the sensing-centric scenario and the tradeoff scenario, it is nonincreasing with a finite lower bound, ensuring the convergence of the proposed algorithm.

Since P1.1, P2.1, P3.1, P4.1, P5.1, and P6.1 are convex problems to be solved using the standard interior point method, the computational complexity of each SCA iteration in Algorithm 1 is denoted as $\mathcal{O}((M+2)^{3.5})$ [46]. Given the convergence precision ϵ, the interior point method requires $\mathcal{O}(\log(1/\epsilon))$ iterations to converge [47]. Let R_{iter} ($R_{\text{iter}} \leq r_{\max}$) represent the number of iterations required by the SCA technique. The overall computational complexity of the SCA technique is then $\mathcal{O}(R_{\text{iter}}(M+2)^{3.5}\log(1/\epsilon))$. In the scenario with K sensing targets, trajectory replanning needs to be executed $(K+1)$ times, resulting in a polynomial upper bound on the computational complexity of Algorithm 1, denoted as $\mathcal{O}((K+1)R_{\text{iter}}(M+2)^{3.5}\log(1/\epsilon))$. It indicates that the proposed algorithm can be solved in polynomial time, making it feasible for practical implementation.

V. Numerical Results

This section validates the performance of our proposed algorithm through extensive numerical experiments. Unless otherwise specified, the experimental scenario involves $N = 5$ CUs and $K = 3$ sensing targets. We further evaluate the overall performance on multiple groups of users and sensing targets in the final experiment. The UAV altitude is set as $H = 100$ m within a working area defined as a $800 \times 800\text{m}^2$ rectangular region. The maximum speed of the UAV is $V_{\max} = 40$m/s and the maximum distance for each trajectory segment is $\Delta_{\max} = 3$m. The starting point and the end point are designated as $\mathbf{q}_I = [0\text{m}, 0\text{m}]^T$ and $\mathbf{q}_F = [800\text{m}, 800\text{m}]^T$, respectively. We set the total mission time as $T = 170$s. Furthermore, the modeling parameters for the probabilistic LoS channel model in (4) are set as $C = 10$, $D = 0.3$, and $k_{\text{LoS}} = 0.98$, $k_{\text{NLoS}} = 0.01$ [32]. The total communication bandwidth is $B = 1$ MHz, and the transmission power of the ISAC signal is $P_T = 20$ dBm. The noise power in the receiver is $\sigma_0^2 = -110$ dBm. The signal processing gain is $G_p = 1$ dB and $G_s = -40$ dB.

The first experiment studies the convergence performance of the proposed algorithm. We observe the convergence properties of the proposed algorithm under the setting $E_{\max} = 25$kJ, $\eta_{\max} = 10$m in the communication-centric scenario, $E_{\max} = 25$kJ, $Q_{\min} = 700$ Mbits in the sensing-centric scenario, and $E_{\max} = 25$kJ, $w_c = w_s = 0.5$ in the tradeoff scenario, respectively. As depicted in Fig. 2, in the three scenarios, the average achievable rate experiences a rapid increase and converges after several iterations using the proposed algorithm. The average CRB demonstrates a rapid descent and stabilizes at a constant value. The results demonstrate that our proposed iterative algorithm based on SCA converges after several iterations in all three typical UAV-enabled ISAC scenarios.

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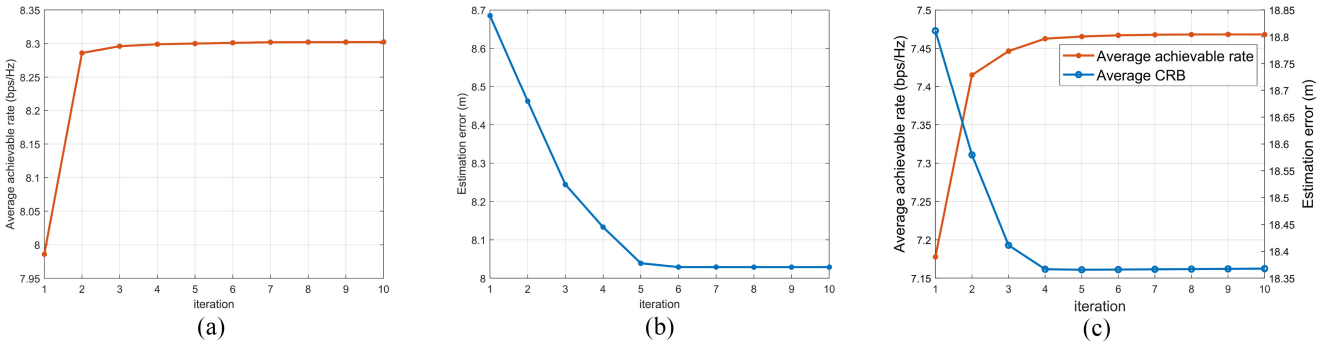


Fig. 2. Convergence behavior of the proposed algorithm in the three scenarios. (a) Communication-centric scenario. (b) Sensing-centric scenario. (c) Tradeoff scenario.

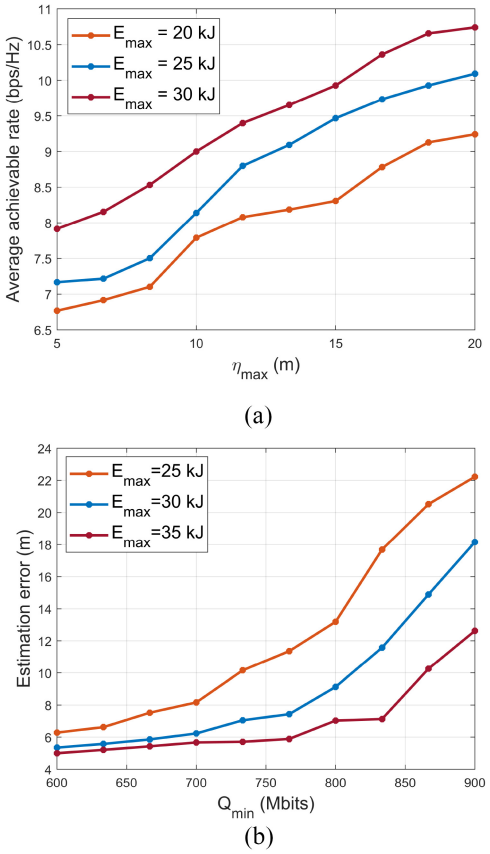


Fig. 3. Impact of different thresholds and energy on performance tradeoff. (a) Communication-centric scenario. (b) Sensing-centric scenario.

This showcases the excellent convergence properties of the proposed algorithm.

The next experiment investigates the tradeoff between communication and sensing performance in typical scenarios. In the communication-centric scenario, different sensing accuracy $\eta_{\max}$ are set to observe the variation in the average achievable rate. As illustrated in Fig. 3(a), with the increasing $\eta_{\max}$, the average achievable rate also increases. This is attributed to the decreasing demand for localization accuracy as $\eta_{\max}$ increases, allowing the system resources to be more inclined toward optimizing communication functionality, thereby enhancing communication performance. Simultaneously, it can be observed that, with the same $\eta_{\min}$ settings, the UAV with higher energy exhibits superior performance. In the sensing-centric scenarios, we observe the variation in the average CRB by adjusting the system communication throughput $Q_{\min}$. As depicted in Fig. 3(b), with the increase in $Q_{\min}$, the average CRB rises, indicating a degradation in the system's localization accuracy. This is attributed to the higher demands on communication throughput, leading to the reallocation of resources from target localization to enhance communication throughput.

The third group of experiments evaluate the impact of changes in UAV onboard energy on communication and sensing performance. As illustrated in Fig. 4, when the maximum available energy for the UAV increases, the system's average achievable rate generally shows an upward trend. At the same time, the system average CRB exhibit an overall decreasing trend. In Fig. 4(a), "com-only" represents the case where only throughput is optimized without sensing constraints, resulting in higher throughput. In Fig. 4(b), "sensing-only" represents the case where only sensing performance is optimized without throughput constraints, resulting in higher localization accuracy. In the tradeoff scenario, the tradeoff between communication and sensing performance is achieved by adjusting the weight w_c and w_s. As w_c increases, system resources tend to prioritize the enhancement of the average achievable rate. This indicates a positive correlation between the network utility of the considered ISAC system and the UAV's energy. With an increase in the UAV energy, the proposed algorithm is capable of adaptively leveraging these resources to enhance both communication quality and target localization accuracy. Fig. 5 shows the impact of different UAV onboard energy on UAV trajectories. Fig. 5(a) shows the UAV trajectories at different onboard energy in the communication-centric scenario with $\eta_{\max} = 5$m. When $E_{\max} = 15$kJ, limited energy keeps the UAV farther from the CUs, reducing the throughput. As energy increases, the UAV flies closer to the CUs, improving the throughput. The UAV also adjusts its trajectory to meet the minimum sensing requirements by choosing appropriate hovering positions and frequencies. Fig. 5(b) shows UAV trajectories in the sensing-centric scenario with $Q_{\min} = 850$ Mbits. Increased energy allows the UAV to extend its trajectory to improve localization accuracy while ensuring minimal throughput by approaching

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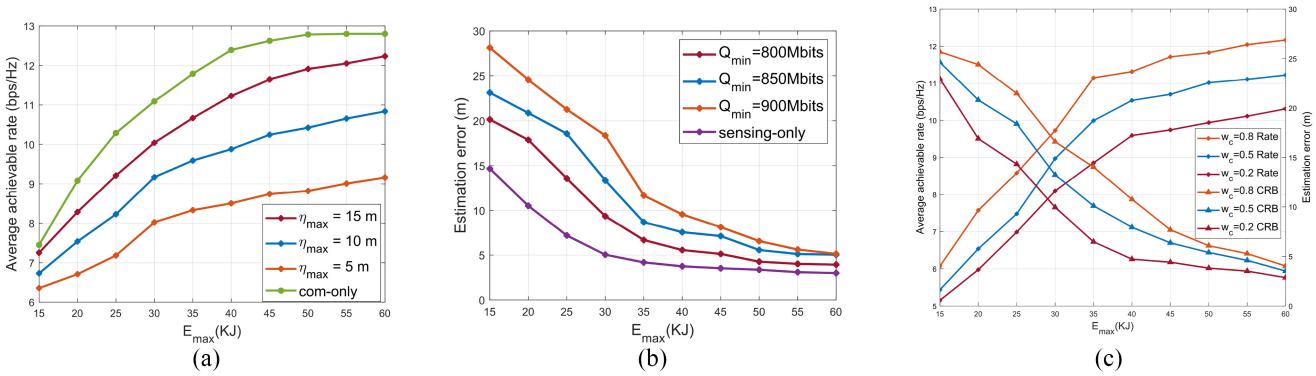


Fig. 4. Impact of different energy on performance in the three scenarios. (a) Communication-centric scenario. (b) Sensing-centric scenario. (c) Tradeoff scenario.

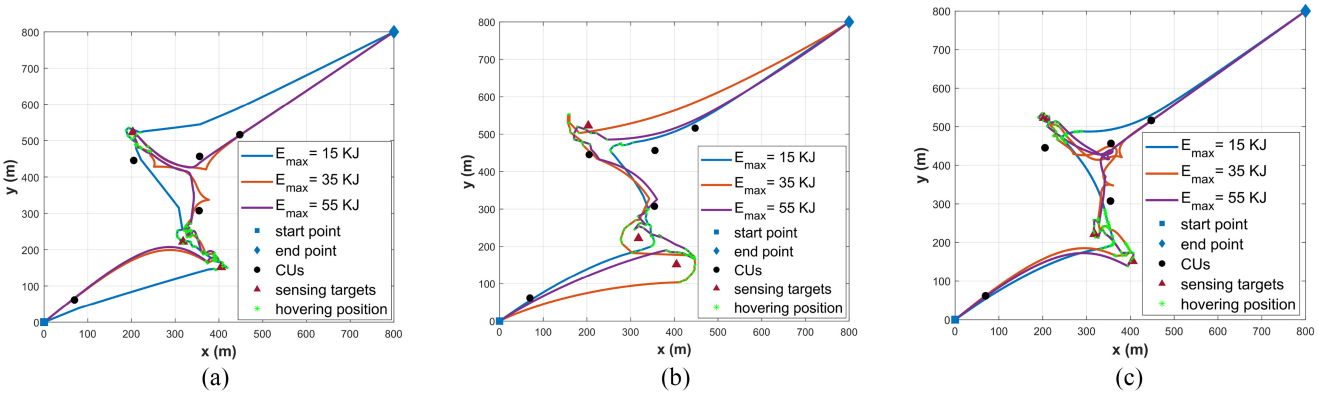


Fig. 5. Impact of different UAV onboard energy on UAV trajectories. (a) Communication-centric scenario. (b) Sensing-centric scenario. (c) Tradeoff scenario.

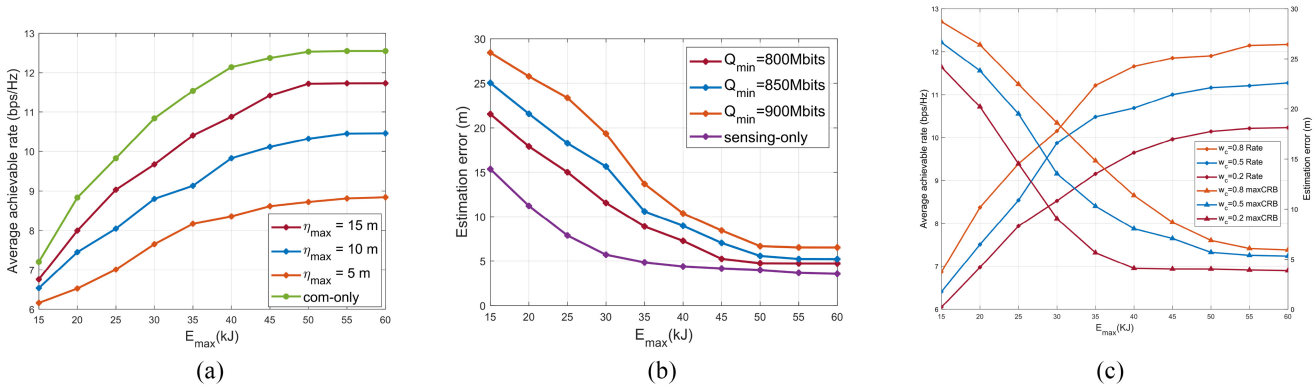


Fig. 6. Impact of different UAV onboard energy on performance under optimization of the maximum CRB. (a) Communication-centric scenario. (b) Sensing-centric scenario. (c) Tradeoff scenario.

CUs. Fig. 5(c) shows UAV trajectories in the tradeoff scenario with $w_c = 0.5$. As onboard energy increases, the UAV moves closer to sensing targets to reduce measurement noise and flies near the optimal communication position of all CUs. This strategy improves throughput, as flying at optimal speed is more energy-efficient than hovering [5].

In Fig. 6, "Estimation error" refers to the maximum CRB of all sensing targets, representing the worst-case localization accuracy of all sensing targets. In the communication-centric scenario, ensuring all targets' CRB are below $\eta_{\max}$ meets sensing constraints, the average achievable rate increases with onboard energy and looser sensing constraints yield higher rates. In the sensing-centric scenario, more onboard energy improves localization accuracy, lowering the maximum CRB. With sufficient energy, all targets achieve high-localization accuracy across different communication constraints. In the tradeoff scenario, higher energy boosts the average achievable rate and reduces all targets' CRB, achieving flexible tradeoffs via w_c and w_s. The results demonstrate that the proposed algorithm can optimize the tradeoff between each target's localization accuracy and network communication throughput based on onboard energy.

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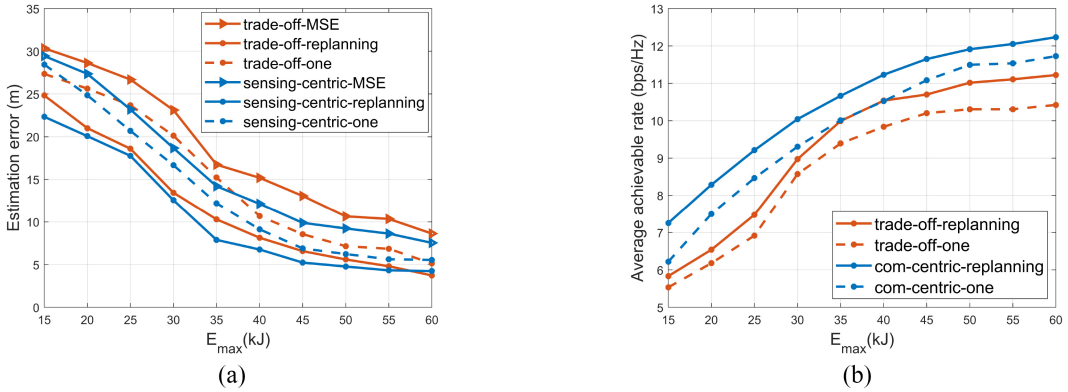


Fig. 7. Impact of different energy on trajectory replanning. (a) Estimation error. (b) Average achievable rate.

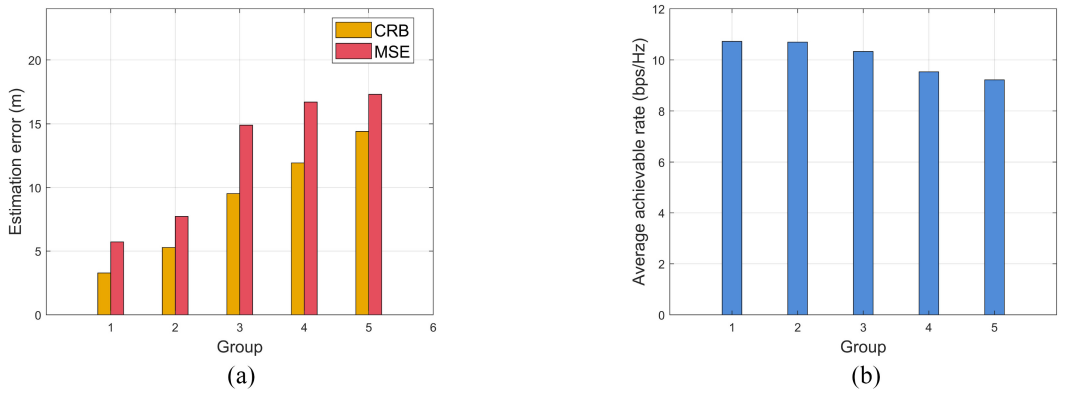


Fig. 8. Performance of scenarios with different number of users. (a) Estimation error. (b) Average achievable rate.

In the next experiment, we present the performance of the proposed trajectory replanning scheme. Without loss of generality, we illustrate the performance comparison between a single trajectory planning and the trajectory replanning algorithm under different energy settings. In Fig. 7(a), we demonstrate the enhancement effect of trajectory replanning scheme on sensing performance. "tradeoff-MSE," "tradeoff-replanning," "tradeoff-one" represent the MSE for single trajectory, average CRB for trajectory replanning and average CRB for single trajectory in the trade-off scenario, respectively. Similarly, "sensing-centric-MSE," "sensing-centric-replanning" and "sensing-centric-one" represent these same metrics in the sensing-centric scenario. In the sensing-centric and tradeoff scenario, setting $Q_{\min} = 850$ Mbits and $w_c = w_s = 0.5$, the trajectory replanning scheme, exhibits an average improvement of 20% in sensing performance. The mean square error of target position estimation is observed to vary with changes in the average CRB, where the average CRB serves as its lower bound. The CRB represents the theoretical lower bound of the actual MSE. Due to limitations, such as sample size, computation errors, and nonideal observation conditions, the actual MSE will be higher than the derived theoretical CRB. In Fig. 7(b), we illustrate the improvement effect of the trajectory replanning scheme on communication performance. "tradeoff-replanning," "tradeoff-one," respectively, represent the average achievable rate for trajectory replanning and single trajectory design in the tradeoff scenario. "com-centric-replanning" and "com-centric-one" represent these same metrics in communication-centric scenario. In the communication-centric and tradeoff scenario, setting $\eta_{\max} = 15$m and $w_c = w_s = 0.5$, the trajectory replanning scheme obtains an average improvement of 5% in communication performance. This indicates that the trajectory replanning scheme, compared with the case of no replanning, facilitates further enhancement of system performance.

TABLE II
NUMBER OF USERS IN GROUP 1 TO GROUP 5

Group	N	K
1	1	1
2	3	2
3	5	3
4	7	4
5	9	6

Finally, we examine the performance of the proposed algorithm in scenarios with different numbers of users. In the experiment, set $E_{\max} = 35$kJ and $w_c = w_s = 0.5$, and the number of users for the five groups are summarized in Table II. In Fig. 8(a), we can see that with the increase in the number of users, the position estimation error of sensing targets gradually increases, because the UAV does not have enough energy to fly closer to the sensing targets. As shown in Fig. 8(b), the average achievable rate in Group 1 and Group

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2 is almost the same, which decreases slightly as the number of users increases in Group 3–Group 5. Because the proposed algorithm adopts the scheme of continuous communication in flight, and the energy consumption of communication is negligible, the system can achieve higher communication throughput when the energy is sufficient, and the increase of the number of users has a minor impact on the throughput.

VI. Conclusion

In this article, we investigated the joint optimization problem of the communication throughput, the targets localization, and the UAV trajectory in the UAV-enabled multiuser ISAC network. First, we derived throughput metrics and the CRB for the position estimation. We used the path discretization model to facilitate flexible design of UAV speed and displacement. Next, we utilized the TSP trajectory initialization based on to determine the UAV service order and the average probability of LoS links. Then, we constructed system problem models for three typical application scenarios of UAV-enabled ISAC networks. For the challenging nonconvex problems, we employed approximation methods to transform the original problems into convex ones. We utilized an iterative algorithm based on SCA to obtain the optimal approximation. Finally, we employed a trajectory replanning algorithm to optimize the UAV trajectory. The numerical results demonstrate that our proposed algorithm effectively addresses the trajectory design problem in a multiuser UAV-enabled ISAC network, achieving an optimal tradeoff between communication and sensing performance.

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Siyan Gu received the B.S. degree in electronic information engineering from Sichuan University, Chengdu, China, in 2022, where she is currently pursuing the M.S. degree with the School of Information and Communication Engineering from University of Electronic Science and Technology of China, China.

Her research interests focus on Internet of Things, integrated sensing and communications, and unmanned aerial vehicle.

Chunbo Luo (Senior Member, IEEE) received the Ph.D. degree in high-performance cooperative wireless networks from the University of Reading, Reading, U.K., in 2011.

His research interests focus on developing model-based and machine learning algorithms to solve networking and engineering problems, with a particular focus on integrated perception, communication, and control design for future autonomous systems. His research has been supported by the National Natural Science Foundation of China, Royal Society, EU H2020, and industries.

Yang Luo received the B.S., M.S., and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2005, 2008, and 2016, respectively.

He is currently an Associate Researcher with UESTC. His research interests include wireless signal processing, the Internet of Things, and machine learning.

Xiaoguang Ma received the bachelor's degree in aerospace engineering and the master's degree in solid mechanics from Beijing University of Aeronautics and Astronautics, Beijing, China, in 1998 and 2001, respectively, and the Ph.D. degree in mechanical engineering from the University of California, Berkeley, Berkeley, CA, USA, in 2005.

He is currently a Professor with the College of Information Science and Engineering and Faculty of Robot Science and Engineering, Northeastern University, Shenyang, China, and Foshan Graduate School, Northeastern University, Foshan, China, where he leads a center focusing on Industrial Artificial Intelligence. His research interests include application of AI on various Industrial areas and unmanned systems.

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